
Lecture Notes

on Compound Events

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1 Compound Event

Def: Let $\mathcal{E}$ be an experiment with sample space $S = \{e_1, e_2, \dots, e_n\}$. We know an event in S is a subset of S . Let E and F be such two events of S , and let e be a particular outcome of the experiment $\mathcal{E}$.

A **compound event** is an event in S that involves some combination of E , F , not E and not F .

$$S, \quad E \subseteq S, \quad F \subseteq S, \quad e \in S$$

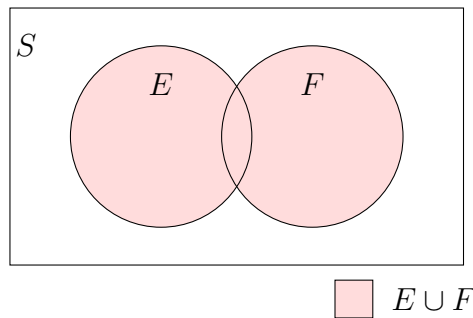
There are **5 different possibilities**:

$e \in E, e \notin E$
 $\nsubseteq$ does not
 $e \in F, e \notin F$

1.1 Event “ E or F ”

The event “ E or F ” occurs if the outcome e is in event E , or in event F , or in both E and F .

This is denoted by $E \cup F$ and read as “ E union F ”.

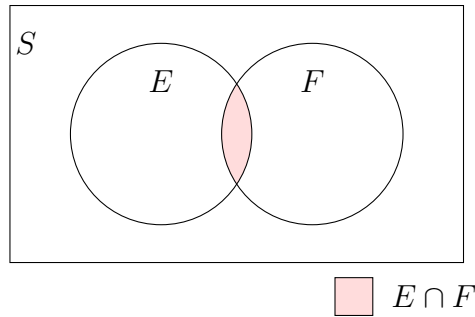


This type of picture is called a **Venn diagram**.

1.2 Event “ E and F ”

The event “ E and F ” occurs if the outcome e is in both event E and event F .

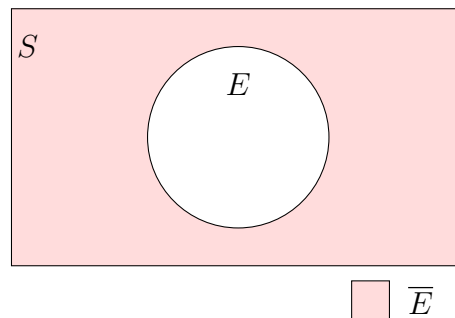
This is denoted by $E \cap F$ and read as “ E intersection F ”.



1.3 Event “Not E ”

The event “not E ” occurs if the outcome e is not in event E .

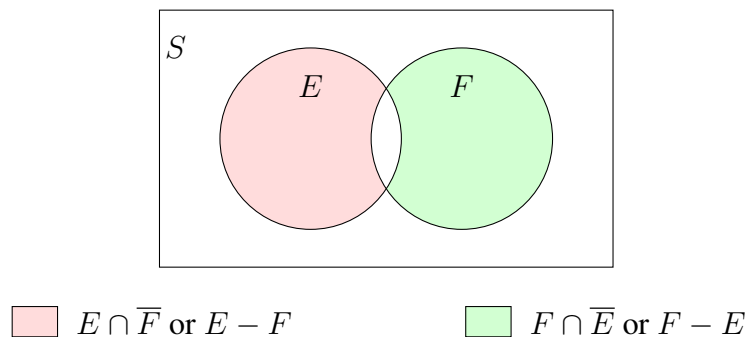
This is denoted by $\overline{E}$ or E' and read as “complement of E ”.



1.4 Event “ E but not F ”

The event “ E but not F ” occurs if the outcome e is in event E but not in event F .

It is denoted by $E \cap \overline{F}$ or $E - F$.



$$P(E \cap \overline{F})$$

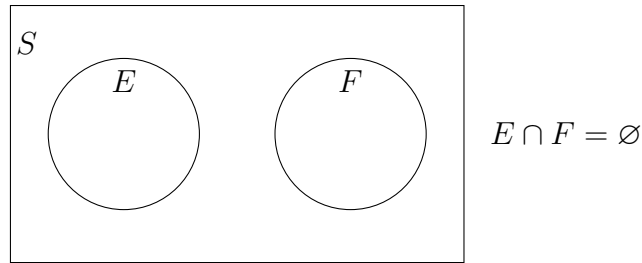
$$P(E - F)$$

$$P(F \cap \overline{E})$$

1.5 Mutually Exclusive Events

We know that $\emptyset$ denotes the **null (empty) set**, which corresponds to an **impossible event**.

Events E and F are called **mutually exclusive** if they cannot happen together, i.e., $E \cap F = \emptyset$.

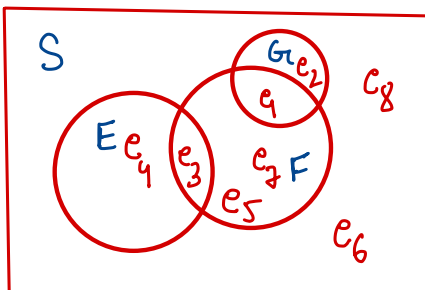


Example: (Blood Type and Rh Factor) Last time, we worked with an example of human blood types A , B , AB , and O . Now we deal with **Rh classification** as an addition.

It is categorized as **positive** and **negative**. You may have heard of blood type B positive and so on.

Q: What is the sample space that consists of blood types with Rh factor?

The sample space S is given by



$S = \{e_1 = \text{Type A with Rh factor +ve},$
 $e_2 = \text{Type A with Rh factor -ve},$
 $e_3 = \text{Type B with Rh factor +ve},$
 $e_4 = \text{Type B with Rh factor -ve},$
 $e_5 = \text{Type AB with Rh factor +ve},$
 $e_6 = \text{Type AB with Rh factor -ve},$
 $e_7 = \text{Type O with Rh factor +ve},$
 $e_8 = \text{Type O with Rh factor -ve}\}.$

$A \rightarrow \begin{matrix} +ve \\ -ve \end{matrix}$
 $B \rightarrow \begin{matrix} +ve \\ -ve \end{matrix}$
 $AB \rightarrow \begin{matrix} +ve \\ -ve \end{matrix}$
 $O \rightarrow \begin{matrix} +ve \\ -ve \end{matrix}$

Let $E = \text{"Type B blood"}$ and $F = \text{"Rh positive"}$.

Q: Find the elementary events of $E \cup F$, $E \cap F$, $\bar{E}$ and $E \cap \bar{F}$.

$E \cup F = \text{Type B or +ve}$

$$E \cup F = \{e_1, e_3, e_4, e_5, e_7\}.$$

$$E \cap F = \{e_3\}$$

$$\bar{E} = \{e_1, e_2, e_5, e_6, e_7, e_8\}$$

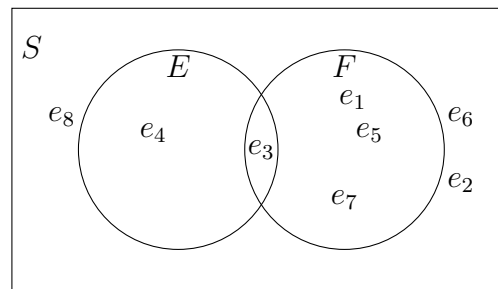
$$E \cap \bar{F} = \{e_4\}.$$

Q: Now, if $G = \text{"Type A blood"}$, check whether E and G are mutually exclusive.

We know that $E \cap G = \emptyset$, since a person cannot have two blood types together. Thus, there is no possible elementary event in $G \cap E$, i.e.,

$$G \cap E = \emptyset.$$

Therefore, E and G are **mutually exclusive events**.



Formulas

$$\left. \begin{aligned} \overline{A \cup B} &= \overline{A} \cap \overline{B} \\ \overline{A \cap B} &= \overline{A} \cup \overline{B} \end{aligned} \right\} \text{De Morgan's Law.}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

2 Probability Axioms

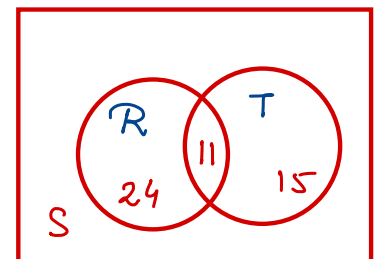
Let S be a sample space. Then:

1. $P(S) = 1$.
2. If E is an event in S , then $0 \leq P(E) \leq 1$.
3. If A and B are **mutually exclusive events** in S (i.e. $A \cap B = \emptyset$), then

$$P(A \cup B) = P(A) + P(B) \quad \text{and} \quad P(A \cap B) = 0.$$

 **Example:** Consider the following information:

Observation	Count
Total cars	39
Red cars	24
Tinted windows	15
Red cars with tinted windows	11



Let R denote the event of a **red car** and T denote the event of a **tinted window**. Find $P(R)$, $P(T)$, $P(R \cap T)$ and $P(R \cup T)$.

$$P(R) = \frac{\text{number of red cars}}{\text{total number of cars}} = \frac{24}{39}.$$

$$\text{Similarly, } P(T) = \frac{15}{39} \text{ \& } P(R \cap T) = \frac{11}{39}.$$

$$\therefore P(R \cup T) = P(R) + P(T) - P(R \cap T)$$

$$= \frac{24}{39} + \frac{15}{39} - \frac{11}{39}$$

$$= \frac{24 + 15 - 11}{39}$$

$$= \frac{28}{39}.$$